

Module 1 Quiz Solutions

Question 1

What fundamental concepts of vector calculus are essential for machine learning optimization?

Answer: B) Gradients, partial derivatives, and directional derivatives

Gradients, partial derivatives, and directional derivatives are core to optimization in machine learning, helping understand how functions change and how to minimize loss functions.

Question 2

When optimizing machine learning models, what is the role of computing gradients?

Answer: B) To understand the curvature of the loss function

Gradients indicate how much the loss function will change with a small change in input values, allowing optimization algorithms to adjust model parameters for better performance.

Question 3

Which optimization algorithm is known for its iterative approach to finding the minimum of a function by moving in the direction of steepest descent?

Answer: B) Gradient descent

Gradient descent is the most common method for optimization, moving iteratively towards the minimum of a function by following the direction of the steepest slope.

Question 4

What is the primary motivation behind using stochastic gradient descent (SGD) instead of traditional gradient descent?

Answer: C) SGD is less computationally intensive

SGD reduces computational complexity by working with mini-batches rather than the full dataset, leading to faster convergence.

Question 5

In machine learning, what does it mean to have an unconstrained optimization problem?

Answer: B) The optimization problem does not involve any constraints on the variables

In unconstrained optimization, the variables are free to take any values that minimize the objective function.

Question 6

Which optimization technique is used to find both local or global minima in unconstrained scenarios?

Answer: C) Simulated annealing

Simulated annealing is a probabilistic method used for finding global minima in complex spaces, making it suitable for unconstrained scenarios.

Question 7

What is the purpose of Lagrange multipliers in constrained optimization problems?

Answer: A) To enforce constraints while optimizing the objective function

Lagrange multipliers allow the inclusion of equality constraints in the optimization process.

Question 8

When dealing with constrained optimization in machine learning, what trade-offs must be considered?

Answer: C) Between computational efficiency and convergence rate

Optimizing while considering constraints requires a balance between finding an optimal solution and the time and resources needed to achieve convergence.

Question 9

Which optimization algorithm moves towards the minimum of a function in discrete steps, making it suitable for large datasets?

Answer: B) Mini-batch SGD

Mini-batch SGD updates the model parameters using small batches of data, combining the advantages of both gradient descent and stochastic gradient descent.

Question 10

Solve the following problem:

$$\min f(x_1, x_2) = \frac{1}{2}(x_1^2 - 2x_2^2) - 2x_1 - x_2$$

Subject to:

$$x_1 + x_2 = 1, \quad x_2 \geq 0$$

Solution Using Karush-Kuhn-Tucker (KKT) Conditions

We define the Lagrangian function:

$$\mathcal{L}(x_1, x_2, \lambda, \mu) = \frac{1}{2}(x_1^2 - 2x_2^2) - 2x_1 - x_2 + \lambda(x_1 + x_2 - 1) + \mu(-x_2)$$

The KKT conditions are:

1. $\frac{\partial \mathcal{L}}{\partial x_1} = x_1 - 2 + \lambda = 0$
2. $\frac{\partial \mathcal{L}}{\partial x_2} = -2x_2 - 1 + \lambda - \mu = 0$
3. $x_1 + x_2 = 1$ (primal feasibility)
4. $x_2 \geq 0 \quad \mu \geq 0 \quad \mu \cdot x_2 = 0$ (complementary slackness)

Case 1: $x_2 > 0$

We get a contradiction, so this case is not valid.

Case 2: $x_2 = 0$

We solve the system to get $x_1 = 1, x_2 = 0$, and the optimal value of the objective function is:

$$f(1, 0) = -\frac{3}{2}$$

Hessian Analysis

The Hessian matrix of the objective function is:

$$H(f) = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$$

The eigenvalues are 1 and -2 , indicating a saddle point at $(x_1, x_2) = (1, 0)$.

Solution Using Substitution

Using the constraint $x_1 + x_2 = 1$, we substitute $x_2 = 1 - x_1$ into the objective function:

$$f(x_1) = -\frac{1}{2}x_1^2 + x_1 - 2$$

The first derivative gives $x_1 = 1$, and substituting back, we find $x_2 = 0$.

The second derivative confirms that the function is concave, and $x_1 = 1, x_2 = 0$ is a maximum.

Thus, the optimal solution is:

$$x_1 = 1, x_2 = 0, \quad f(1, 0) = -\frac{3}{2}$$